

2.5

Storing memories

How do we retain memories in the long term?

Our LTerm capacity is essentially limitless

by studying nu connections, researchers estimated its storage capacity as "in the same ballpark as the World Wide Web"

memories do not reside in specific spots

researchers removed various parts of rats' brains' cortex, no matter which section removed, rats retained atleast partial mem of how to navigate the maze

mems are brain based, but brain distributes the components of a mem across network of locations.

these locations incl. some of the nu circuitry involved in the earlier experience

Mem is not like a library w/ precise locations.

What is the biological basis for memory?

Explicit mems either { Semantic - facts general knowledge
Episodic - experienced events

Semantic networks connect concepts - helping us group objects based on characteristics they share.

We more readily store new ExMems if they fit within existing schemas.

Regions that process & store new ExMems incl.
Frontal lobes & hippocampus.

When you remember past exp., many brain regions send input to prefrontal cortex, for working memory processing

Left Frontal lobes processes verbal & factual info in mems
Right Frontal lobes processes visual & spatial info in mems

Hippocampus, a temporal lobe structure in limbic system, helps process exmems.

As children grow → hippocampus matures, enabling construction of more detailed mems

damage to Hpc disrupts formation & recall of exmems

Left Hpc damage → trouble w/ verbal info
but no trouble with visual designs & locations

Right Hpc damage → reverse problem

rear of hippocampus processes spatial memory &
it grows bigger, ex. london cabbies learn to navigate the city

mems not permanently stored in Hpc

brain registers & temp. holds elements of an episode, smell, feel, sound & location in Hpc

then sends mems to cortex for storage

this process is called memory consolidation.

Sleep supports memory consolidation

In deep sleep, Hpc processes mem for later retrieval

↑ Hpc activity in sleep $\Rightarrow$ ↑ next day's mem

Hpc & brain cortex displayed simultaneous activity rhythms during sleep, as if they were having a dialogue.

Brain seems to replay day's experiences as it transfers them.

Personal Connection

I've recently switched from Quizlet to Anki, which is designed around distributed learning (and its not evil) to improve recall

Learning is distributed $\rightarrow$ more sleep

$\rightarrow$ more chance for mem consolidation

What role do cerebellum & basal ganglia play in memory processing?

Cerebellum plays key role in forming & storing imMem created by classical conditioning.

Basal Ganglia, deep brain structures involved in motor movement, facilitate formation of our procedural memories for skills

BG receive input from the cortex, but don't send info back to cortex for conscious awareness of procedural learning

This is why skills learned during infancy persist

As adults, our conscious memory of first 4 years is largely blank :: infantile amnesia

2 reasons:

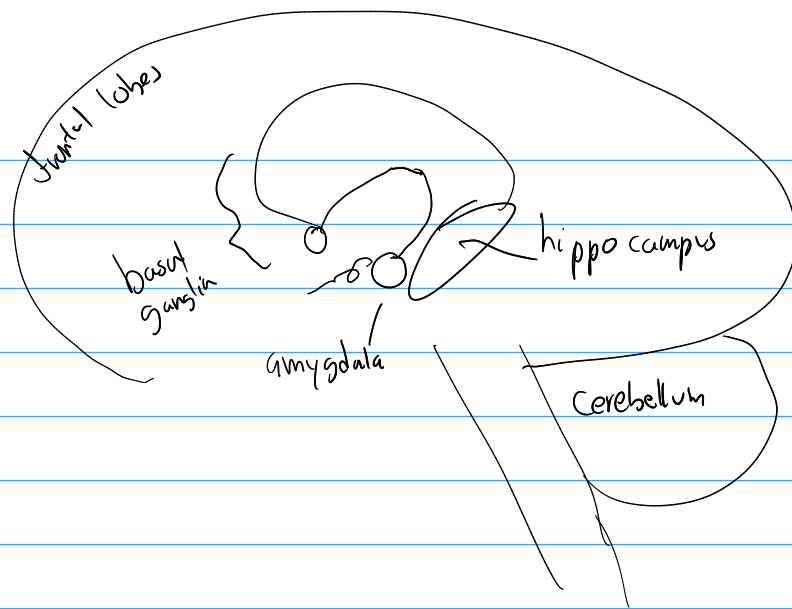
- we index much of exmem thru command of language that young children do not have
- Hpc one of last brain structures to mature

How do emotions affect memory processing?

emotions trigger stress hormones that influence mem formation

When we're excited or stressed, these hormones make more glucose energy available to fuel brain activity, signaling to brain something important is happening

Stress provokes amygdala to initiate a memory trace that boosts activity in brain's memory-forming areas



very stressful events can form **unforgettable** memories

emotional events produce **tunnel-vision** memory:
focus attention & recall on high-priority information
reduce recall of irrelevant details

Flashbulb memories are **clear mental snapshots** of exciting or shocking events.

95% of american adults said they could recall exactly where they were & what they were doing when they heard of 9/11

Over time, even flashbulb memories may become inconsistent

MCQ:

1.c 2.c 3.b 4.b 5.a